

User Experiences with Chronic-Disease Self-Management Apps: A Qualitative Analysis of Google Play and Apple App Store Reviews

Boyan Tong

I. Introduction

Chronic diseases, formally known as chronic noncommunicable diseases (CNCDs), include cardiovascular diseases (such as heart disease and stroke), cancer, chronic respiratory diseases (such as chronic obstructive pulmonary disease and asthma), and diabetes (World Health Organization, 2025). In 2021, at least 43 million deaths were caused by noncommunicable diseases, accounting for approximately 75% of all non-pandemic-related deaths worldwide (World Health Organization, 2025). In the United States, about three-quarters of adults have at least one chronic disease, and more than half of adults have two or more chronic diseases (U.S. Centers for Disease Control and Prevention, 2026a). Therefore, chronic diseases are not merely a problem faced by a small number of patients but rather a long-standing and far-reaching public health issue.

Currently, the management of chronic diseases is an integral part of patients' daily lives. Patients with chronic diseases must take medication as prescribed and monitor health indicators such as blood glucose or blood pressure; in addition to clinical interventions, patients must also cope with daily symptoms—such as pain, fatigue, or sleep disturbances

—as well as psychological stress (CDC, 2026b). Mobile health applications play a significant role in the management of chronic diseases by digitizing self-management tasks to reduce the burden on patients. Specifically, these applications can help patients record and track symptoms and health indicators, remind them to take medications or attend appointments, provide health information and personalized guidance, facilitate communication with healthcare professionals, and support remote monitoring and consultations. They may also identify abnormal health trends and issue timely alerts.

Although there are a large number of mobile health apps currently on the market, many articles point out shortcomings in the apps themselves and in existing research on them. Regarding mobile health apps, Iribarren et al. (2021) note that mobile health apps effects on improving health are relatively limited. Similarly, Cheah et al. (2024) point out that many apps fail to meet development standards and lack the involvement of healthcare professionals. Some apps offer only standardized reminders and basic data-logging functions, making it difficult for them to adapt to the complex and ever-changing daily lives of patients with chronic diseases. In addition, research on chronic disease apps often faces challenges such as short follow-up periods and small sample sizes (Iribarren et al., 2021).

For chronic disease self-management apps, only a few studies have used application store reviews. For instance, Haggag et al. (2022) conducted a large-scale classification of over 5 million reviews for 278 mHealth applications. However, their analysis focused on engineering issues such as stability, privacy, and interface. Therefore, the existing research still rarely starts with patients' natural usage reviews to compare valued features and current usability and accessibility barriers across major types of chronic diseases in a single study.

Based on this research gap, the study aims to address the following research questions. RQ1: What features do reviewers value in chronic-disease self-management apps on Google Play and the Apple App Store? RQ2: What usability and accessibility barriers do reviewers report when using these apps?

II. Related Work

1. Understanding How and Why Patients with Chronic Diseases Use Self-Management Apps

1.1 Monitoring and Recording Health Data

Patients with chronic diseases typically record their symptoms, medication, diet, and daily activities at specific times to track trends in

their data and support their treatment. Schroeder et al. (2018) conducted a study on the self-tracking behaviors of migraine patients. They found that patients recorded data to identify potential triggers of their condition, evaluate treatment effectiveness, and prepare for medical appointments. However, intermittent symptoms and incomplete records often make it difficult to interpret the data. Vafeiadou et al. (2021), in a study on self-tracking among people with Parkinson's disease, further noted that patients recorded symptoms, medication responses, and daily activities to analyze the relationships among them. Burford et al. (2019) directly examined how patients with type 2 diabetes use data visualization for self-management. They found that records and trend graphs of blood glucose, diet, weight, and other metrics can help patients review their behaviors before and after abnormal readings occur and consider adjusting their dietary plans accordingly. A study by Huniche et al. (2013) on the use of self-monitored readings by patients with chronic obstructive pulmonary disease (COPD) showed that these readings can help patients decide whether to adjust their activity levels or contact healthcare providers.

However, the monitoring capabilities provided by self-management apps do not equate to clinical accuracy. Coppetti et al. (2017) compared the results of four heart rate monitoring apps with those from electrocardiograms (ECGs) and pulse oximeters, finding significant

variations in accuracy among the different apps. Kumar et al. (2015) conducted a content analysis of 107 hypertension apps and found that some claimed to measure blood pressure directly without using professional measurement tools such as cuffs. Plante et al. (2016) subsequently conducted an independent clinical evaluation of a popular app. They found that the app incorrectly classified 77.5% of participants with hypertension as non-hypertensive, with a sensitivity of only 0.22 for identifying hypertension. However, the same team's study of reviews for this app revealed that users still frequently gave it high ratings and described it as "accurate," rarely recognizing the actual errors (Plante et al., 2018). Therefore, in my analysis, I must always be mindful of the subjectivity of user reviews, which cannot replace objective evidence from medical experiments.

1.2 Seeking Personalized Support

The key to personalized support is providing the help patients truly need at the right time and in the right way. Nahum-Shani et al. (2018) noted in their design framework for real-time adaptive interventions that the system must determine whether to provide support, when to provide it, and what type of support to offer based on the user's current state and context. Refraining from intervention when users do not need or are unable to receive it is, in itself, part of personalization.

Personalization in self-management apps includes timed reminder features. Stawarz et al. (2014) evaluated 229 medication reminder apps and analyzed 1,012 user reviews, finding that reminders relying solely on fixed timers often overlook the relationship between medication-taking behavior and daily routines. McBride et al. (2020) had patients with hypertension use blood pressure tracking and medication reminder features continuously for four weeks and found that reminders primarily helped patients who were prone to unintentionally missing doses, but they might disrupt established routines. Therefore, reminder features should allow users to adjust the time, frequency, sound, pause periods, and trigger conditions; users should not be treated as having identical daily routines.

Personalization also involves tailoring the interface, functions, and information to the characteristics of an individual's chronic illness and individual preferences. In a study on the app needs of patients with cystic fibrosis, Hilliard et al. (2014) found that patients wanted to reduce repetitive data entry through automation, system integration, and customization features. They were not satisfied with a single reminder function, and their attitudes toward gamification and reward mechanisms were inconsistent. Korpershoek et al. (2020), in a user-centered design study on self-management of acute exacerbations of chronic obstructive pulmonary disease (COPD), found that patients preferred clear,

uncluttered interfaces, and some considered cartoon characters to be childish. Alwashmi et al. (2020) conducted direct interviews with COPD patients and healthcare professionals. The app feature requirements identified by respondents included disease education, symptom assessment and metric tracking, connectivity with external devices, family access, and doctor-patient communication.

1.3 Sharing Health Information with Others

Another core function of self-management apps is recording and sharing data with family members and doctors, and this often includes using tables and trend charts to transform personal data into information that doctors can quickly understand. Chung et al. (2015) directly studied how healthcare professionals use patients' lifestyle log data and found that this data can facilitate communication and personalized recommendations. However, issues such as limited clinical time, a lack of data standardization, and export methods that do not align with workflows can hinder data utilization. Schroeder et al. (2017) designed visualization tools centered on food and symptom logs from patients with irritable bowel syndrome, enabling patients and healthcare providers to interpret the data together. Reviewing the data collaboratively helps both parties understand the information, build trust, and arrive at actionable decisions. In a study of self-tracking data during

clinical visits for patients with Parkinson's disease, Mentis et al. (2017) also demonstrated that graphical representations can tangibly influence treatment discussions between patients and clinicians. Therefore, apps should not only provide sharing functionality but also support summaries with selectable time ranges, trend charts, anomaly flags, notes, and export formats optimized for clinical review.

The recipients and timing of data sharing also require clarification and specification within the app. A study by Lawton et al. (2021) on data sharing among adolescents with diabetes and their parents found that real-time access to data can support parents in providing remote care but may also cause stress and conflict depending on the child's age and family dynamics. The study therefore recommends establishing agreements in advance regarding who can view which data and when interventions should occur. A study by Hilliard et al. (2014) on cystic fibrosis also showed that patients both wanted apps to help them connect with their healthcare teams and were concerned about privacy risks associated with the centralized storage and sharing of disease data. After conducting a meta-analysis of 21 diabetes apps and 602 reviews, Turner and Stawarz (2025) found that connection failures, inaccessibility, malfunctioning notifications, software updates, and hardware compatibility issues can all disrupt the sharing process. Consequently, apps need to support real-time sharing features for family

members while clearly defining access permissions and providing a revocation function.

2. Design and Evaluation of Self-Management Apps for Chronic Diseases

2.1 Identifying the Specific Needs of Patients with Different Chronic Diseases

A self-management app is typically designed for patients with a single type of chronic disease and must offer a high degree of customization to accommodate the needs of different individuals. Ayobi et al. (2021) summarized an 18-month co-creation study on self-care technology for type 1 diabetes, noting that patients' needs regarding insulin, blood glucose, and diet are highly individualized. Beyond diabetes, cystic fibrosis patients want apps that automate repetitive tasks, support prescription refills, facilitate communication with healthcare teams, and provide disease-specific information (Hilliard et al., 2014). In contrast, patients with chronic obstructive pulmonary disease (COPD) place greater emphasis on identifying acute exacerbations, connecting to pulse oximeters or pulmonary function devices, and implementing personalized action plans (Korpershoek et al., 2020; Alwashmi et al., 2020). These studies all focus primarily on disease-specific daily tasks, illustrating that design requirements should be grounded in disease management processes.

Cancer and cardiovascular diseases also exhibit distinct information and decision-making needs. Taramasco et al. (2023), through a co-creation study of a symptom-reporting app involving breast cancer patients and healthcare providers, identified needs such as information on the disease and treatment, guidance on daily living, and explanations of medical administrative processes and costs; they also noted that interface design must account for the emotional burden associated with disease-related information. When using apps, patients with hypertension prioritize blood pressure trends, abnormal changes, medication reminders, and communication regarding medical appointments (McBride et al., 2020). Therefore, the specific needs of patients with different chronic diseases vary significantly. For example, diabetes management focuses on tracking blood glucose, insulin, and carbohydrate intake; cardiovascular disease management emphasizes blood pressure or heart rate monitoring, trend analysis, and measurement thresholds; and chronic respiratory disease management prioritizes symptom changes, medication adherence, and acute exacerbations.

2.2 Enhancing Features to Support Self-Management Tasks

Regarding the features of most chronic disease self-management apps, Donevant et al. (2018), in their review of app features and outcomes for

chronic respiratory diseases, diabetes, and hypertension, identified nine categories of features—including tracking, reminders, uploading physiological metrics, personalized goals, two-way communication, and clinical decision support—based on 70 original studies. A rapid review of chronic disease app features by Gomolin et al. (2020) found that reducing page transitions, manual input, and authentication steps can improve app usability. The feature set proposed by Alwashmi et al. (2020) for COPD also indicates that disease education, metric monitoring, action plans, medication reminders, and communication are only meaningful when centered on the actual tasks of patients and healthcare providers.

For recurring data-entry tasks, reliability, usability, and data recording are core functions. Turner and Stawarz (2025) found that failures in device connectivity, data access, notifications, and software updates in diabetes apps increase the workload of re-entering data and compromise safety. He et al. (2024) conducted a comparative study of user satisfaction based on 261 hypertension apps and 116,686 reviews from China and the United States. They found that usability, blood pressure tracking, and interface design contribute to satisfaction, while factors such as lack of synchronization, reliability, compatibility, measurement accuracy, cost, and advertising often lead to dissatisfaction.

2.3 Evaluating App Accessibility and Usability

Usability evaluation requires observing whether target users can complete real-world tasks; it cannot rely solely on star ratings. In a systematic review of 656 diabetes apps, Arnhold et al. (2014) conducted expert usability testing on 66 apps with patients aged 50 and older and found that the more features an app had, the lower its overall usability tended to be. Apart from screen readers, accessibility features such as font size enlargement and color adjustments are relatively limited in 29 of 29 iOS apps and 11 of 27 Android apps. Sarkar et al. (2016) had patients and caregivers from diverse backgrounds test 11 commercial health apps. Out of 185 tasks, only 79 could be completed without assistance, with the main difficulties stemming from extensive manual input, multi-layered pages, and unclear navigation. Yu et al. (2019) directly compared the performance of users with limited hand dexterity using the original and updated versions of a system, finding that text prompts, adjustable text and buttons, customizable keyboards, and shorter operation paths improved efficiency and usability. Therefore, font, contrast, information density, operational steps, and error recovery should all be considered as independent dimensions of analysis.

The long-term user experience also depends on technical stability, service quality, business model, and privacy. Through focus groups and

interviews with 38 patients, Giebel et al. (2024) categorized patient concerns into ten categories: effectiveness, usability, technology, usage and adherence, privacy and security, doctor-patient relationships, knowledge and skills, individual differences, implementation, and cost. These concerns included crashes, update or compatibility issues, low willingness to pay, and concerns about advertising. Haggag et al. (2022) conducted a large-scale analysis of over 5.1 million reviews of 278 mobile health apps and found that account login, payments and subscriptions, stability, updates, compatibility, connectivity, privacy, and developers' failure to respond to feature requests were all issues repeatedly reported by users. Building on these findings, I will pay particular attention to the usability and accessibility deficiencies mentioned in the articles.

III. Methods

This study analyzed reviews of chronic disease self-management apps on Google Play and the Apple App Store to understand the features users valued and the accessibility issues they encountered. These two platforms were selected because, as the largest mainstream mobile app stores, they cover Android and iOS apps, respectively. Therefore,

including both platforms reduces the bias that would result from studying only a single platform.

User reviews were selected because they capture authentic, direct feedback from users after actual use and better reflect the app's functionality and issues in real-world scenarios than the app descriptions written by developers.

1. Data Collection

The raw data for this study was sourced from English-language user reviews on Google Play and the Apple App Store. Based on the World Health Organization's classification of the four major categories of chronic noncommunicable diseases, which are cardiovascular diseases, cancer, chronic respiratory diseases and diabetes. The keywords for each disease category were first defined based on the World Health Organization's classification of the four major categories of noncommunicable diseases and then formulated by combining official disease names with commonly used aliases or abbreviations. The keywords were as follows. Cardiovascular diseases: "heart failure," "congestive heart failure," "hypertension," and "high blood pressure"; cancer: "cancer," "oncology," and "cancer care"; chronic respiratory diseases: "asthma," "COPD," and "chronic obstructive pulmonary

disease"; diabetes: "diabetes," "type 1 diabetes," "type 2 diabetes," and "blood glucose."

Searches in both app stores were conducted in the United States account in July 2026. The top five apps returned for each search term were recorded, and duplicate apps were consolidated. The researcher then screened the apps based on app name, store description, and target audience. Eligible apps must address one of the target diseases, be intended for patients or caregivers, provide at least one self-management feature, have English-language reviews, and remain available for download at the time of data collection. Excluded apps included tools intended for healthcare professionals, general health or fitness apps, apps offering only one-time screenings, and apps with no clear relationship to the target disease. From the eligible apps, two were selected for each disease category, for a total of up to eight apps, with priority given to apps available on both platforms and those with a large number of reviews.

For each app, the most recent 30 English-language text reviews were collected separately from each platform. Researcher used ChatGPT Sol 5.6 to assist in writing two Python-based web scraping scripts to batch-retrieve reviews based on the app's Google Play package ID or Apple App ID. The scripts exported the review text, star rating, review date,

app name, platform, and corresponding ID into Microsoft Excel. The scraping results were manually verified by researcher. Developer replies, duplicate reviews, advertisements, non-English reviews, and reviews without text ratings were excluded. To ensure confidentiality, reviewers' usernames, profile pictures, and other identifying information were not collected.

The final coding data included 137 reviews. The cardiovascular disease category included 29 reviews for InPulse - Heart Rate Monitor and 15 reviews for Blood Pressure Tracker SmartBP. The diabetes category included 14 reviews for mySugr - Diabetes Tracker Log and 30 reviews for Gluroo Diabetes Logger. The chronic respiratory diseases category included 14 reviews for Asthma Tracker and 9 reviews for Aluna - Lung Health Management. The cancer category included 12 reviews for Outcomes4Me Cancer Care and 14 reviews for LivingWith™: Cancer Support.

2. Data Analysis

2.1 App and Review Characteristics

Descriptive statistics were used to report the apps' disease categories, primary functions, prices, and platform distribution, as well as the number of reviews, star rating distribution, and average ratings across

each platform and disease category. Coding categories are not mutually exclusive; therefore, a single review may be counted under multiple themes.

2.2 Qualitative Analysis

Reviews underwent inductive coding using qualitative content analysis. The analysis consisted of three phases: familiarization with the data, theme formation through inductive coding, and interpretation of the research findings. Valued features and usability and accessibility barriers were analyzed after finishing the inductive coding.

The researcher randomly selected 35 of 137 reviews from the 8 included apps, representing approximately 25.5% of the review dataset, for pilot inductive coding. Coder used Skimle to assist with identifying key meaning units and grouping similar concepts into Level 2 codes, which were reviewed by the coder as well. The coder then manually consolidated Level 1 codes into Level 1 codes and then themes. The codebook documented the definition of each code along with 1–3 examples from the original reviews, and codes were merged, deleted, or modified based on the results of the pilot coding.

The final coding followed the same coding process as pilot coding. And the final coding framework adopted a three-level structure consisting of

Themes, Level 1 codes, and Level 2 codes. Among these, Level 2 is closest to the original text and summarizes specific comments. Level 1 groups together Level 2 codes with similar meanings. Themes integrate related Level 1 codes into analytical dimensions that may address the research questions. For example, one reviewer wrote, "My Apple Watch immediately recorded my heart rate jumping up to 154 BPM, while this app continued to show 83 BPM." The researchers first identified "the app's readings did not reflect the heart rate changes shown by the reference device" as a meaning unit. This excerpt was coded as Level 2: "The app failed to reflect the heart-rate change shown by a reference device." It was subsequently classified as Level 1: "Reports fluctuations in heart-rate readings or discrepancies with a reference device, thereby casting doubt on accuracy." The final assigned Theme was "Functional Performance and Measurement Accuracy."

Once the final coding framework was established, the coder analyzed the remaining reviews one by one and re-examined the pilot coding sample using the final framework.

IV. Findings

1.1 Cardiovascular Diseases: Pulse Measurement and Personalized Blood Pressure Management

There were two self-management apps related to cardiovascular diseases: InPulse and SmartBP.

Of the 29 reviews for InPulse, 12 addressed the accuracy of heart rate or pulse measurements; 4 were positive, while 8 raised concerns. One user, after comparing the app's results with their own measurements, wrote, "I've counted it myself then done it in the app and have had the exact same number." This user also noted, "The app is so much easier than counting my pulse for 15 seconds and multiplying by 4." However, there were also negative reviews from users who obtained conflicting results when using other monitoring devices. For example, one user who simultaneously measured their heart rate with an Apple Watch stated, "My Apple Watch immediately recorded my heart rate jumping up to 154 BPM, while this app continued to show 83 BPM." This demonstrated that being able to measure their pulse anytime using a smartphone provided some users with convenience and peace of mind. However, users' trust in this feature was conditional. Whenever readings clearly conflict with those from auxiliary devices or their own physical sensations, they became skeptical of the feature's accuracy.

The blood pressure measurement feature was mentioned in six reviews, two of which were from SmartBP and concerned custom tags and adjustable blood pressure reference ranges; the other four were from InPulse and related to blood pressure measurement or recording

functions. One user wrote, “The flexibility of the tagging features in this app allows you to tag & track—then graph—whatever you want: stress levels, salty diet, ... specific medications or supplements, caffeine intake.” Users could record changes in stress, diet, and medications, then observe the relationship between these factors and their blood pressure. These features integrated blood pressure records with individual treatment contexts, helping users understand which changes warrant attention and increasing their motivation to continue tracking. Another user highlighted the ability to adjust blood pressure thresholds based on personal circumstances—specifically, modifying the blood pressure range with a doctor’s guidance so that it “reflects the parameters that apply to me.”

1.2 Diabetes: Dose Tables, IOB Calculations, Carb Management, and Hypoglycemia Alerts

Of the 44 reviews for diabetes self-management apps, 10 fell under the “Blood Glucose Monitoring and Diabetes Management Features” subcategory. Of these, 8 were for Gluroo and 2 for mySugr. Nine reviews generally praised the core features, while one discussed technical

issues.

The features of greatest interest to diabetes users centered on blood glucose, insulin on board, and carbohydrate content calculations. One user particularly praised the status page for displaying "estimated carbs / insulin on board," noting that even hours after a meal or insulin administration, it still "gives a rough estimate of what has been metabolized and what is left in the system." Another user particularly praised the app's dosing table and insulin-on-board calculations: "I love the dosing table and the insulin-on-board calculations to help me make decisions. And I am very well-versed in insulin calibration." For users who needed to frequently factor in past doses, meals, and current blood glucose levels, these centralized calculation features were especially useful. Other users also commended the app's ability to issue predictive alerts before blood glucose levels dropped too low. One user, Sugarmate, commented, "If I don't hear the low glucose alert, I get a phone call telling me I'm low." Other users also praised the app's reminders for replacing consumables. One patient mentioned that the app displayed whether the insulin pump was in "automated/limited/manual" mode.

1.3 Chronic Respiratory Diseases: FEV1 Logs

There were two self-management apps related to chronic respiratory

diseases: Aluna and Asthma Tracker. Six comments specifically addressed features related to chronic respiratory diseases: three for Aluna—one of which praised the daily FEV1 trend tracking, while two reported an inability to complete measurements when readings were low or symptoms worsened—and three for Asthma Tracker, all of which noted that measurements could only be recorded once per day and that multiple peak flow readings within a single day could not be saved.

Forced Expiratory Volume in One Second (FEV1) refers to the volume of air exhaled during the first second of a forced exhalation. One Aluna user wrote, "My numbers usually drop the day before I have an asthma attack. Because of Aluna, I've been able to prevent most of my bad asthma days." The user stated that, based on the app's historical FEV1 data, they would take preventive inhalers before their numbers dropped and always carry a rescue inhaler.

However, users of both apps complained that FEV1 recording failures often occurred during periods of worsening symptoms, making it impossible to capture data when breathing was most difficult. A user who described themselves as having moderate COPD wrote, "The app doesn't seem to like it when your levels are below 60 or 50 ... and often says 'try again.'" The user also stated, "60+ percent of the tests I do, I don't pass." A comment regarding severe asthma noted, "It [is] not able to record any score on days when my asthma is bad." The user further

explained that after blowing into the device, “the rocket ship spins and a measurement can’t be taken. It can’t take a measurement on days when I need it most.”

Additionally, users often needed to measure FEV1 multiple times per session to more accurately predict when an asthma attack would occur. Among three comments on Asthma Tracker regarding the frequency of repeat entries, one parent wrote that their child “has problems in the morning and then does great in the evening, but the app only lets you make one entry per day.” Another user requested the addition of “day and night, or multiple entries for one day” to record multiple peak flow readings throughout the day.

1.4 Cancer: Personalized Treatment Plans, Clinical Trial Screening, and Expert Consultations

There were two cancer-related self-management apps: Outcomes4Me and LivingWith. Reviews of Outcomes4Me primarily focused on personalized disease information and expert medical consultations, whereas reviews of LivingWith more frequently addressed coordination among family members and other supporters. Therefore, the reviews for LivingWith fell under the common features highlighted by users discussed later in this section. Among the 12 reviews for Outcomes4Me, three fell under the “Specific Disease Information and Specialized

Features” subcategory, covering personalized treatment plans based on treatment history and disease progression, recommendations for potentially relevant clinical trials, and the inability to enter multiple cancer diagnoses simultaneously.

The primary challenge faced by cancer patients is the complexity of treatment information; consequently, they prioritize features that filter information based on treatment history and disease progression. One Outcomes4Me user wrote, “This app offers specific information on treatment options tailored to your treatment history and disease progression... it suggests current studies for which you are eligible... [They] offer one-on-one consultations with oncology nurses.” Another user summarized this combination as follows: “From understanding side effects to accessing second opinions and clinical trial matches, it’s like having a supportive, super-smart advocate in your pocket.” It is evident that users appreciated the app’s ability to recommend treatment information, clinical trials, and nurse consultations that aligned with their disease progression.

1.5 Commonly Cited Features

User feedback primarily concerned the interpretation of medical information and data, the visualization of long-term trends, and the exchange of information between different devices and healthcare

providers. There were 8 comments on “layman’s medical explanations,” coming from Outcomes4Me and SmartBP; there were 14 comments on “statistical page visualization,” distributed across 5 apps, of which 13 were positive and 1 was mixed; “Data Synchronization and Sharing” received a total of 12 comments across 4 apps, including 10 positive comments, 1 negative comment, and 1 mixed review.

1.5.1 Enhancing Medical Literacy and Reducing the Cognitive Burden

Users expressed interest in features that translated specialized medical content or large volumes of health data into easily understandable information, including lay-person explanations of pathology reports, prompts for questions to ask during consultations, and color-coded representations of the severity of measurement results.

An Outcomes4Me user recalled their experience upon diagnosis: “I was so lost when I was first diagnosed with cancer. The app helped me make sense of my pathology report, understand my treatment options, and even brought up questions I hadn’t thought to ask my doctor.” Another user added, “They really try to explain things in a way that’s understandable to the layperson.” The app reduced aimless searching by patients and alleviated the confusion they faced when seeking specialized information after a diagnosis. SmartBP offered another

approach: it used “color coding of the measurements” to indicate the “HBP severity” corresponding to blood pressure readings and provided links to health education resources.

1.5.2 Visualization on the Statistics Page

Presenting trend changes through charts is a feature that has garnered significant attention in user feedback. A mySugr user stated, “I can see the trend of my levels and know at a glance that I’m okay.” A SmartBP user added, “Having the graphs and stats is encouraging when I see the downward trend.” Both comments affirmed that trend graphs help users quickly view changes and can even lift their spirits.

However, user comments also pointed out limitations in chart presentation. A SmartBP user noted that while the app’s screen uses colors to distinguish blood pressure ranges, historical reports “are generated in black and white,” making it “impossible to glean anything from a quick glance.” An Aluna user observed, “The history by week shows my score of 67 ... however, the history by month completely leaves out the score of 67.”

1.5.3 Data Synchronization and Sharing

Users of various chronic disease apps need to share health information across different devices, with family members, and with doctors.

Users first mentioned inviting family members to view the health data of children or the elderly. One Gluroo user explained that they use the app “to bridge my CGM app to share my readings with my spouse,” after which their partner can “look on his phone to know how I’m doing.” Here, CGM refers to continuous glucose monitoring devices. One parent noted that when their child attended a six-week summer camp, the information provided by the app—such as IOB, device mode, and phone battery level —“was everything I needed to feel at ease and communicate with the nurse,” enabling the parent and nurse to collaborate based on the same set of real-time information.

Some users also value the app’s ability to automatically import data from external devices. For example, people with diabetes value the ability to automatically import data via HealthKit or CGM and view their current carbohydrate intake and IOB on the same screen: “The app can pull my HealthKit data in... to show my current carb / IOB.”

Other users also appreciate the feature that allows them to send detailed PDF charts to their doctors to identify patterns and discuss treatment plans. The export feature generates color PDFs that can be forwarded directly to doctors via email. One reviewer of a blood pressure app

stated, "Sharing/export options are great both for data backup and for providing BP logs to doctors."

Negative and mixed reviews, however, mentioned that the Apple Watch requires indirect connection via other apps, that Libre data import is incomplete, and that data cannot be automatically synchronized.

2. Usability and Accessibility Barriers

User feedback on usability and accessibility barriers primarily involves three categories. The first category involves font sizes that are too small and prompts that disappear too quickly, which hinder normal use by older adults. The second category concerns fixed measurement frequencies and times that fail to accommodate the fluctuating symptom patterns of patients with chronic conditions. The third category involves interruptions in long-term record-keeping caused by updates, paywalls, and device integration issues. The fourth category relates to a lack of credibility in readings or health messaging. The anger, disappointment, and willingness to abandon the app expressed by users reflect how these usage bottlenecks negatively impact their health management.

2.1 Older Adults: Too-Small Font Sizes and Briefly Displayed Prompts

The subtopic "Visual Design and Accessibility" received a total of 11 comments, 8 of which were negative and related to SmartBP, mySugr,

and Gluroo. These issues included font sizes that were too small and prompts that disappeared too quickly.

A SmartBP user who identified as an older adult complained that the font size was too small: "The letters and numbers are minuscule and impossible for me to see without a magnifying glass." A Gluroo user first praised the lock screen chart as "perfect & easy to grasp at a glance," but then requested, "PLEASE make the IOB bigger & easy to see from a dead sleep without glasses or a magnifying glass." Another user requested that the prompt be displayed for a longer period: "I finally figured out that if I time the screen capture just right, I can read it that way, but I'm an old man—how coordinated am I supposed to be?" Since the apps may lack font size adjustment and do not allow customization of prompts, they are extremely unfriendly to the elderly population.

2.2 Patients with Chronic Conditions: Lack of Flexibility in Measurement Frequency and Timing

Eight separate comments related to this topic concerned the three apps Asthma Tracker, mySugr, and Aluna. Six of these pointed out that fixed recording frequencies, data volume requirements, or timing rules fail to accommodate users' symptom fluctuations throughout the day and their existing measurement schedules. The other two comments noted that when breathing is most difficult—and when measurements are most

needed and critical—users are often required to retake them or are unable to generate results.

Three reviews of Asthma Tracker noted that users can only log data once per day. Two reviews from mySugr users pointed out that a fixed number of daily measurements is required to obtain an A1C estimate. One mySugr user wrote, “It requires me to test 3 times a day for 7 consecutive days,” but noted that they only measure twice daily due to their existing schedule. Another user described how the app required a measurement one hour before a meal, but they were still preparing breakfast at that time: “I didn’t start eating until just before 9 a.m. It said I failed the challenge.” This user felt that the system “doesn’t make accommodations for meal preparation.” A review on LivingWith noted, “ALL of the help sections are already selected by default.”

In a mySugr review, a user wrote: “No dark mode?! Diabetics often have low vision as well.” Impaired vision is a common complication of diabetes, so the lack of a dark mode directly impacts the target users’ ability to use the app normally. Additionally, a user named Aluna wrote, “The app doesn’t seem to like it when your levels are below 60 or 50. I have moderate COPD, and it doesn’t like my numbers—it often tells me to ‘try again’... I fail over 60 percent of the tests I take, and it asks me to retake them.” This user has reduced lung function, yet the app frequently flags the results as failing the test and requires a retest.

Repeated testing increases the physical burden and may make it difficult to accurately record more severe lung conditions.

2.3 Updates and Payment Models Disrupt Long-Term Self-Management

"Subscription Pricing and Business Models" involved 24 comments across 6 apps, of which 14 were negative, 9 were positive, and 1 was neutral. Another 7 comments explicitly linked the issue to updates or version changes, involving 5 apps and all 4 disease categories.

Changes to payment models may disrupt users' health data, which is particularly detrimental to low-income and vulnerable populations. One InPulse user wrote, "All this basic but super important info that I've been tracking... is locked behind a paywall that I can't afford. Important notes that I was going to show my doctor [were] completely blocked off. No warning either." A user who identified as disabled wrote, "Being disabled has made money so tight, and one of my worries is that this app will become too expensive for me to use." Clearly, paywalls limit the number of times core features can be used, block access to medical records, and even cause frequent ad pop-ups. The high cost severely restricts access for low-income patients with chronic conditions.

At the same time, software updates may render core features inoperable, including tracking metrics, accessing history, and sharing content. A LivingWith user bluntly pointed out that the sharing feature is

restricted: "Since this last update, I can't seem to figure out how to share my posts with them." A mySugr user also noted that the data visualization feature had disappeared: "This leaves you guessing where you might be trending toward your A1C goal... since the DMI feature is no longer available, I've got no idea where I'm heading... Very disappointed with no projected date for an impending 'bug fix' in sight." Clearly, for users who rely on longitudinal data to monitor chronic conditions, these updates severely disrupt the continuity of medical records and hinder communication between patients and healthcare providers.

2.4 Unreliable and Misleading Feature Claims

Questions regarding measurement reliability and product descriptions were clearly concentrated on InPulse. Of the 12 reviews regarding heart rate accuracy, 4 affirmed the readings, while 8 raised doubts.

Some users proactively cross-checked the app's readings with other devices. One InPulse user who could not tolerate standing described the experience: "I followed the instructions and launched the heart rate app on my Apple Watch at the same time... I decided to stand up while measuring my heart rate... my Apple Watch immediately recorded my heart rate jumping to 154 BPM, while this app continued to show 83 BPM." InPulse's failure to reflect this obvious increase in heart rate cast doubt on the accuracy of the app's claimed heart rate monitoring

capabilities. One user wrote, "InPulse assigns a heartbeat or BPM to anything that is red." Users even tested the app with non-human red objects—such as vases—in front of the camera, yet the app still detected a heart rate. Some even suspected that the app lacks actual heart rate monitoring capabilities and merely returns a random "measurement" based on the intensity of red pixels within a certain range. Many genuine users subsequently lost confidence in this core feature.

The description of the blood pressure feature sparked similar controversy. Another InPulse user highlighted the gap between the promotional language and the actual functionality, and wrote, "The app doesn't monitor your blood pressure. You may measure your own blood pressure physically ... Then you may choose to enter your data manually." This user suggested using terms like "track," "journal," or "log" to describe this feature. For health apps, the terms "measure" and "record" create different user expectations. Misleading product descriptions may lead users to misjudge the monitoring capabilities of the phone itself.

V. Discussion

I believe the most significant insight from the Findings section is that, for chronic disease self-management apps, accessibility barriers can

profoundly and negatively impact patients' ability to use core features, thereby complicating or even hindering the treatment process. Furthermore, since patients' ability to operate the app diminishes as their symptoms worsen, accessibility barriers may arise precisely when patients most need to record changes in their physical condition. For example, low vision caused by diabetes can make it difficult to read data, while repeated measurements become a burden when lung capacity declines.

The impact of accessibility barriers also accumulates over time with prolonged use. Repeated failures to read the interface or complete tests can result in missing entries, leading to incomplete statistical data and potentially increasing the workload for patients and caregivers. If the app consistently rejects values deemed "unreasonable," long-term records may also omit data from periods when the condition is more severe. Therefore, evaluations of chronic disease apps must incorporate usage scenarios during symptom flare-ups and periods of reduced physical ability, as static interface ratings or task tests conducted while the user is in good health may underestimate the actual barriers.

Based on the Findings section, particularly the discussion on accessibility barriers, here are some recommendations for patients, developers, and platforms:

1. Recommendations for Patients

Before downloading a self-management app, users should thoroughly read the product feature descriptions and user reviews to identify misleading claims early on, understand which core features are restricted by paywalls, and gain insight into the actual user experience. A good self-management app typically stores and integrates medical records, generates long-term trends based on physiological data, and offers sharing and multi-user login capabilities. However, when it comes to medication dosages or emergencies, patients should contact healthcare professionals directly, as these apps cannot replace professional medical advice.

Users who need family collaboration can configure viewing permissions and reminders for trusted family members. Patients can also discuss key metrics and personal thresholds with their doctors to ensure the app's records are better suited for follow-up care communication.

2. Recommendations for Developers

① Developers could consider conducting co-design sessions focused on specific conditions. Participants should represent users of different ages and varying disease severities. Testing could consider also consider including usage scenarios during symptom exacerbations—for

example, whether core tasks can be completed during episodes of low lung function, nocturnal hypoglycemia, or impaired vision.

② Accessibility design must cover the entire management process. Dynamic fonts, high contrast, dark mode, screen reader support, large touch areas, and simplified input could apply consistently to logs, charts, reminders, and reports.

③ Product marketing could consider clearly defining product features, distinguishing between measurement, estimation, and recording. Features involving physiological signals must specify validation methods, error margins, supported devices, and the types of medical judgments for which the product is not intended. Developers could consider also monitor whether users are relying on supplementary information for formal medical decisions and provide prominent warnings in such cases.

④ Product continuity could be considered as a design requirement. Updates must preserve historical data; provide stable data migration and rollback options; and notify users before core features are removed, reminder mechanisms are changed, or subscription terms are adjusted.

⑤ Developers could also clearly specify the commercial use of patient data in their terms of service to maximize the protection of patient privacy.

3. Recommendations for Platform Providers

① The Apple App Store and Google Play could consider exercise greater caution in controlling the listing, promotion, recommendation, and pricing of health apps and assume greater responsibility for platform governance. Platforms can establish stricter submission requirements for apps that claim to measure, predict, or monitor physiological indicators, requiring developers to explain the measurement principles and the intended use of patient data to prevent false advertising and ensure patient privacy is not compromised.

② Health app pages could include standardized information labels displaying clinical validation, privacy policies, accessibility features, compatible devices, data export methods, and core pricing. Search and recommendation algorithms could also incorporate this information to reduce reliance on star ratings and download counts.

③ Platforms also need to improve transparency in comment moderation, strengthen the review of a large number of suspicious accounts that appear within a short period of time and scrutinize positive or negative comments with similar content. When comments are deleted, the platform could specify the rules on which the decision is based and retain internal audit records, thereby mitigating the impact of developers manipulating comments or users posting malicious reviews.

VI. Limitations

1. Authenticity of Reviews Cannot Be Verified

App store reviews may contain paid reviews and bot-generated content. In the InPulse dataset, some reviewers noted that “Most of these reviews very obviously seem botted”; others accused developers of “botting their reviews and deleting real ones.” At the same time, we are unable to verify the true identities of the reviewers making these accusations. This study cannot confirm whether each raw review comes from a real patient and reflects a genuine user experience.

For future research, I suggest that researchers preemptively identify and exclude text that is identical or highly similar and examine reviews that appear in clusters over a short period of time, as well as accounts with unusual names.

2. Reviewers Do Not Represent the Majority

App store reviews come from people who actively choose to post them. Users who have encountered serious malfunctions, are extremely satisfied, or are strongly dissatisfied with the cost may be more willing to publicly share their experiences. We cannot obtain feedback from the silent majority.

3. Reviewer Identities Cannot Be Verified

We cannot verify the identities of reviewers through public comments. Whether the reviewer is a patient, a family member, a healthcare professional, or a short- or long-term user, these factors influence the perspective and depth of the review. We also lack background information such as the reviewer's age, disease severity, and digital literacy. Therefore, self-reported information cannot be verified against medical records.

Anonymity also limits our ability to interpret differences in user experiences. For example, font issues may be related to device screen size or personal preferences, and failed tests may be influenced by user error during device operation. However, the study cannot distinguish between these factors through follow-up questioning.

4. Interpretation of Themes Is Influenced by the Researcher's Judgment

Coding is inherently a highly subjective task, and the fact that this study was conducted by a single coder further exacerbates bias. In future studies, it is recommended that two or more coders be assigned to enhance the rigor of the research as follows: After the coders complete the initial coding, they will randomly sample 20% of the full dataset and recode it without referring to the original coding results. Each code will

be converted into a binary variable based on its presence or absence, and Cohen's kappa will be calculated between the two coding rounds to assess intra-coder consistency. If the kappa for a major code falls below 0.70, the coders will re-examine its definition and review the relevant comments using the revised coding framework.

5. Language and Regional Limitations

This study analyzed only English-language reviews on the U.S. versions of Google Play and the Apple App Store. However, user habits, payment experiences, and app features may be influenced by language, regional policies, and healthcare systems; therefore, the study's findings cannot be directly generalized to other countries or language groups.

VII. Conclusion

This study analyzed self-management apps for four major chronic diseases: cardiovascular disease, diabetes, chronic respiratory diseases, and cancer. The review revealed that apps designed for different diseases tend to feature distinct core functionalities. Diabetes users prioritize features such as dosage tracking, IOB, and carbohydrate management. Users with respiratory diseases prioritize lung function

measurements and early warning systems for exacerbations. Users with cardiovascular disease are highly concerned with upper and lower limits of blood pressure readings. Cancer users prioritize personalized treatment information, clinical trials, and expert consultations. Cross-disease reviews also revealed common needs. Users want medical information to be easy to understand, to observe long-term trends through charts, and to share data with family members or doctors.

Negative user reviews primarily focused on issues regarding the reliability of readings and promotional claims. Some users also encountered barriers related to font size and device connectivity. Updates, feature removals, and changes to pricing models may result in historical data or core features becoming unavailable.

By coding users reviews, the main contribution of this study lies in its comparative analysis of the specific and common features of disease-specific apps, as well as barriers to accessibility. The findings of this study can help patients make more informed choices when selecting apps and assist developers in continuously improving and testing core app features. App platforms can use these findings as a reference to strengthen their review processes for health app feature information.

Reference

Alwashmi, M.F., Fitzpatrick, B., Davis, E., Farrell, J., Gamble, J.-M. and Hawboldt, J. (2020) 'Features of a mobile health intervention to manage chronic obstructive pulmonary disease: a qualitative study', *Therapeutic Advances in Respiratory Disease*, 14, 1753466620951044. doi:10.1177/1753466620951044.

Arnhold, M., Quade, M. and Kirch, W. (2014) 'Mobile applications for diabetics: a systematic review and expert-based usability evaluation considering the special requirements of diabetes patients age 50 years or older', *Journal of Medical Internet Research*, 16(4), e104. doi:10.2196/jmir.2968.

Ayobi, A., Stawarz, K., Katz, D., Marshall, P., Yamagata, T., Santos-Rodriguez, R., Flach, P.A. and O'Kane, A.A. (2021) 'Co-designing personal health? Multidisciplinary benefits and challenges in informing diabetes self-care technologies', *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2), article 475, pp. 1–26. doi:10.1145/3479601.

Burford, S.J., Park, S. and Dawda, P. (2019) 'Small data and its visualization for diabetes self-management: qualitative study', *JMIR Diabetes*, 4(3), e10324. doi:10.2196/10324.

Centers for Disease Control and Prevention (2026a) About chronic diseases. Available at: CDC: About chronic diseases (Accessed: 16 August 2026).

Centers for Disease Control and Prevention (2026b) Living with a chronic condition. Available at: CDC: Living with a chronic condition (Accessed: 16 August 2026).

Cheah, K.J., Abdul Manaf, Z., Mat Ludin, A.F., Razalli, N.H., Mohd Mokhtar, N. and Md Ali, S.H. (2024) 'Mobile apps for common noncommunicable disease management: systematic search in app stores and evaluation using the Mobile App Rating Scale', *JMIR mHealth and uHealth*, 12, e49055. doi:10.2196/49055.

Chung, C.-F., Cook, J., Bales, E., Zia, J. and Munson, S.A. (2015) 'More than telemonitoring: health provider use and nonuse of life-log data in irritable bowel syndrome and weight management', *Journal of Medical Internet Research*, 17(8), e203. doi:10.2196/jmir.4364.

Coppetti, T., Brauchlin, A., Müggler, S., Attinger-Toller, A., Templin, C., Schönraht, F., Hellermann, J., Lüscher, T.F., Biaggi, P. and Wyss, C.A. (2017) 'Accuracy of smartphone apps for heart rate measurement', *European Journal of Preventive Cardiology*, 24(12), pp. 1287–1293. doi:10.1177/2047487317702044.

Donevant, S.B., Estrada, R.D., Culley, J.M., Habing, B. and Adams, S.A. (2018) 'Exploring app features with outcomes in mHealth studies involving chronic respiratory diseases, diabetes, and hypertension: a targeted exploration of the literature', *Journal of the American Medical Informatics Association*, 25(10), pp. 1407–1418. doi:10.1093/jamia/ocy104.

Giebel, G.D., Abels, C., Plescher, F., Speckemeier, C., Schrader, N.F., Borchers, K., Wasem, J., Neusser, S. and Blase, N. (2024) 'Problems and barriers related to the use of mHealth apps from the perspective of patients: focus group and interview study', *Journal of Medical Internet Research*, 26, e49982. doi:10.2196/49982.

Gomolin, A., Lebouché, B., Engler, K. and Vedel, I. (2020) 'Optimizing smartphone intervention features to improve chronic disease management: a rapid review', *Health Informatics Journal*, 26(3), pp. 1795–1809. doi:10.1177/1460458219891377.

Haggag, O., Grundy, J., Abdelrazek, M. and Haggag, S. (2022) 'A large scale analysis of mHealth app user reviews', *Empirical Software Engineering*, 27(7), article 196. doi:10.1007/s10664-022-10222-6.

He, Y., Zhu, W., Wang, T., Chen, H., Xin, J., Liu, Y., Lei, J. and Liang, J. (2024) 'Mining user reviews from hypertension management mobile health apps to explore factors influencing user satisfaction and their asymmetry: comparative study', *JMIR mHealth and uHealth*, 12, e55199. doi:10.2196/55199.

Hilliard, M.E., Hahn, A., Ridge, A.K., Eakin, M.N. and Riekert, K.A. (2014) 'User preferences and design recommendations for an mHealth app to promote cystic fibrosis self-management', *JMIR mHealth and uHealth*, 2(4), e44. doi:10.2196/mhealth.3599.

Hunich, L., Dinesen, B., Nielsen, C., Grann, O. and Toft, E. (2013) 'Patients' use of self-monitored readings for managing everyday life with COPD: a qualitative study', *Telemedicine and e-Health*, 19(5), pp. 396–402. doi:10.1089/tmj.2012.0135.

Iribarren, S.J., Akande, T.O., Kamp, K.J., Barry, D., Kader, Y.G. and Suelzer, E. (2021) 'Effectiveness of mobile apps to promote health and manage disease: systematic review and meta-analysis of randomized controlled trials', *JMIR mHealth and uHealth*, 9(1), e21563. doi:10.2196/21563.

Korpershoek, Y.J.G., Hermesen, S., Schoonhoven, L., Schuurmans, M.J. and Trappenburg, J.C.A. (2020) 'User-centered design of a mobile health intervention to enhance exacerbation-related self-management in patients with chronic obstructive pulmonary disease (Copilot): mixed methods study', *Journal of Medical Internet Research*, 22(6), e15449. doi:10.2196/15449.

Kumar, N., Khunger, M., Gupta, A. and Garg, N. (2015) 'A content analysis of smartphone-based applications for hypertension management', *Journal of the American Society of Hypertension*, 9(2), pp. 130–136. doi:10.1016/j.jash.2014.12.001.

Lawton, J., Hart, R.I., Kimbell, B., Allen, J.M., Besser, R.E.J., Boughton, C., Elleri, D., Fuchs, J., Ghatak, A., Randell, T., Thankamony, A., Trevelyan, N., Hovorka, R. and Rankin, D., on behalf of the CLOuD Consortium (2021) 'Data sharing while using a closed-loop system: qualitative study of adolescents' and parents' experiences and views', *Diabetes Technology & Therapeutics*, 23(7), pp. 500–507. doi:10.1089/dia.2020.0637.

McBride, C.M., Morrissey, E.C. and Molloy, G.J. (2020) 'Patients' experiences of using smartphone apps to support self-management and improve medication adherence in hypertension: qualitative study', *JMIR mHealth and uHealth*, 8(10), e17470. doi:10.2196/17470.

Mentis, H.M., Komlodi, A., Schrader, K., Phipps, M., Gruber-Baldini, A., Yarbrough, K. and Shulman, L. (2017) 'Crafting a view of self-tracking data in the clinical visit',

in Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems. New York: Association for Computing Machinery, pp. 5800–5812. doi:10.1145/3025453.3025589.

Nahum-Shani, I., Smith, S.N., Spring, B.J., Collins, L.M., Witkiewitz, K., Tewari, A. and Murphy, S.A. (2018) ‘Just-in-time adaptive interventions (JITAI) in mobile health: key components and design principles for ongoing health behavior support’, *Annals of Behavioral Medicine*, 52(6), pp. 446–462. doi:10.1007/s12160-016-9830-8.

Plante, T.B., Urrea, B., MacFarlane, Z.T., Blumenthal, R.S., Miller, E.R. III, Appel, L.J. and Martin, S.S. (2016) ‘Validation of the Instant Blood Pressure smartphone app’, *JAMA Internal Medicine*, 176(5), pp. 700–702. doi:10.1001/jamainternmed.2016.0157.

Plante, T.B., O’Kelly, A.C., Macfarlane, Z.T., Urrea, B., Appel, L.J., Miller, E.R. III, Blumenthal, R.S. and Martin, S.S. (2018) ‘Trends in user ratings and reviews of a popular yet inaccurate blood pressure-measuring smartphone app’, *Journal of the American Medical Informatics Association*, 25(8), pp. 1074–1079. doi:10.1093/jamia/ocy060.

Sarkar, U., Gourley, G.I., Lyles, C.R., Tieu, L., Clarity, C., Newmark, L., Singh, K. and Bates, D.W. (2016) ‘Usability of commercially available mobile applications for diverse patients’, *Journal of General Internal Medicine*, 31(12), pp. 1417–1426. doi:10.1007/s11606-016-3771-6.

Schroeder, J., Hoffswell, J., Chung, C.-F., Fogarty, J., Munson, S. and Zia, J. (2017) ‘Supporting patient-provider collaboration to identify individual triggers using food and symptom journals’, in Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing. New York: Association for Computing Machinery, pp. 1726–1739. doi:10.1145/2998181.2998276.

Schroeder, J., Chung, C.-F., Epstein, D.A., Karkar, R., Parsons, A., Murinova, N., Fogarty, J. and Munson, S.A. (2018) ‘Examining self-tracking by people with migraine: goals, needs, and opportunities in a chronic health condition’, in Proceedings of the 2018 Designing Interactive Systems Conference. New York: Association for Computing Machinery, pp. 135–148. doi:10.1145/3196709.3196738.

Stawarz, K., Cox, A.L. and Blandford, A. (2014) ‘Don’t forget your pill! Designing effective medication reminder apps that support users’ daily routines’, in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. New York: Association for Computing Machinery, pp. 2269–2278. doi:10.1145/2556288.2557079.

Taramasco, C., Rimassa, C., Noël, R., Bravo Storm, M.L. and Sánchez, C. (2023) 'Co-design of a mobile app for engaging breast cancer patients in reporting health experiences: qualitative case study', *Journal of Medical Internet Research*, 25, e45968. doi:10.2196/45968.

Turner, J. and Stawarz, K. (2025) 'Understanding device integrations within diabetes apps: mixed methods analysis of app features and user reviews', *JMIR Diabetes*, 10, e62926. doi:10.2196/62926.

Vafeiadou, J., Vasalou, A. and Roussos, G. (2021) 'Self-tracking in Parkinson's: the lived efforts of self-management', *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), article 84, pp. 1–25. doi:10.1145/3449158.

World Health Organization (2025) Noncommunicable diseases, 25 September. Available at: WHO: Noncommunicable diseases (Accessed: 16 August 2026).

Yu, D., Parmanto, B. and Dicianno, B. (2019) 'An mHealth app for users with dexterity impairments: accessibility study', *JMIR mHealth and uHealth*, 7(1), e202. doi:10.2196/mhealth.9931.